Mathematics Qiyuan Ma

Research catalogue

Locally irregular colourings of doubled graphs

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001. One manuscript, one problem family. The catalogue number is a local navigation label. The internal A1–E4 groups collect related assertions; they are not a count of separate papers or independently new theorems.

The problem

For a finite simple graph G, let ${}^2G$ replace each edge by two parallel copies. Each copy receives red or blue. Whenever a colour occurs on an edge, its two endpoints must have different degrees in that colour. The original conjecture asks for two colours for every connected $G \ncong K_2$.

Principal result: the subcubic case

The manuscript presents a computer-assisted proof of

$$G \text{ finite, connected, simple,} \quad G \ncong K_2, \quad \Delta(G) \le 3 \quad \Longrightarrow \quad \operatorname{lir}({}^2G) \le 2.$$

The proof combines structural reductions, prescribed local interfaces, complete finite kernels and lifting arguments.

Further sufficient conditions

For simple G, the paper also gives the following two sufficient conditions:

$$\delta(G) \ge 8\chi(G)(2\chi(G)+1) - 7, \qquad 8\delta(G) \ge 7|V(G)| + 6.$$

Here δ is minimum degree and χ is chromatic number. Under the second condition, the red degrees are distinct on all vertices and the blue degrees are distinct on all vertices. The hypotheses and small-order scope are exactly those in the full manuscript.

Constructions and interfaces

Further sections address sparse branch cores, balanced subdivisions, clique root targets and block-tree capacity, preserved colour counts, dense forbidden targets, complement lifts, and signed-gap and parity-port constructions. Explicit examples describe obstacles to specified stronger strategies. They are not counterexamples to the original conjecture.

Scope. The unrestricted conjecture for arbitrary connected graphs is not proved. This is an AI-assisted research preprint, without independent expert peer review or proof-assistant verification claimed.

Materials and verification

Read and build

The complete 64-page manuscript, its sixteen TeX source files, bibliography, build instructions and BibTeX citation are in the preprint directory. Run `tectonic main.tex` there, or run `pdflatex main.tex` twice. The original PDF is unchanged by this catalogue update.

Complete finite proof data

The directly browsable A1 subset contains five JSON certificates and six verifier modules. It covers tree/cactus ports, sparse cores, two-marked and marked-theta interfaces, fresh local checks and rejection of three deliberately damaged certificates. The weak-two-marked kernel has 4,256 cases.

The complete snapshot

The frozen archive contains all 697 released files, including constructors, other complete certificates, examples, provenance and previous reports. Additional checks cover 836 clique bases, four E1–E4 auditors recounting 972 saved colour copies, and 1,114 bounded endpoint cases supporting the displayed general path argument. These different checking layers have different scopes.

Standard-library commands, from the repository root:

```
python -I -B proof_artifacts/subcubic/verify.py
python -I -B verify_snapshot.py
```

Python 3.10 or later is required. These commands need no third-party packages, network, solver or account. Do not use `-O`. Constructors have separate dependency and scope instructions in the archive.

What the checks mean

Complete finite calculations verify their specified domains and witnesses. Bounded or random tests check only their tested inputs. The infinite-family conclusions also depend on the manuscript's structural and lifting arguments. Python certificate checking is not Lean formalisation.

Provenance, novelty and revisions

The source scope is frozen at 10:25:33 CST on 28 October 2026. OpenAI Codex assisted with exploration, proof development and checking, programming, literature screening and writing. The novelty note distinguishes candidate contributions from known special cases and classical tools; no absolute-priority certificate is claimed. Corrections and versions are recorded in the public history.

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